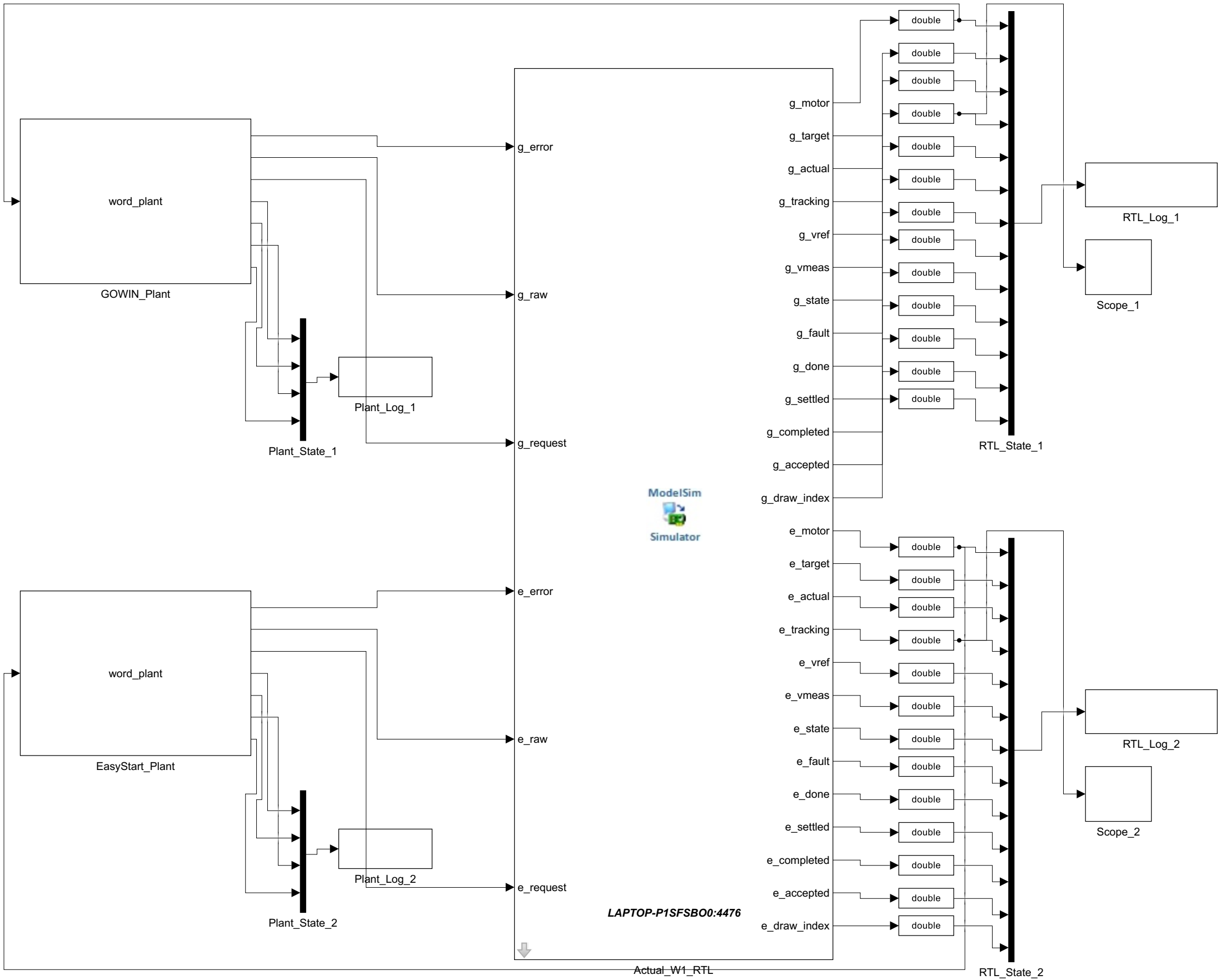


Actual C5 RTL in ModelSim;5ms interface,one-sample delay;50MHz computation edges;idle clocks omitted.

Supplied Oct27 representative count-domain plant;not identified IP570/Furuta hardware;no ideal encoder following.

Supplied plant states: x=[angle error; angle rate; arm position; arm velocity].
Five semi-implicit 1 ms steps per 5 ms sample: a=80*x1+6*u-0.5*x2; aa=-0.02*u-x4.
x2+=0.001*a; x1+=0.001*x2; x4+=0.001*aa; x3+=0.001*x4. Initial x=[12;0;0;0].
u is the actual RTL motor request. Original-kit raw encoder=round(x3*27/26); RTL maps 26/27.